

Composición de funciones

Determinar $f \circ g$, $g \circ f$, $f \circ f$ y $g \circ g$

Para $f(x) = 3 - 2x$; $g(x) = 6 - 3x$, calcular

- $f(3a)$: $f(3a) = 3 - 2a$
- $f(\frac{3}{a})$: $f(\frac{3}{a}) = 3 - 2(\frac{3}{a})$;
- $f(\frac{1}{2})$: $f(\frac{1}{2}) = 3 - 2(\frac{1}{2})$
- $f(\frac{1}{2}x)$: $f(\frac{1}{2}x) = 3 - 2(\frac{1}{2}x) = 3 - x$
- $f(-3x)$: $f(-3x) = 3 - 2(-3x) = 3 + 6x$
- $f(6 - 3x)$: $f(6 - 3x) = 3 - 2(6 - 3x) = 3 - 12 + 6x = -9 + 6x$
- $f(g(x))$: $f(g(x)) = f(6 - 3x) = 3 - 2(6 - 3x) = -9 + 6x$
- $f(f(x))$: $f(f(x)) = f(3 - 2x) = 3 - 2(3 - 2x) = 3 - 6 + 4x = -3 + 4x$

$$\text{Dominio}(f) = \mathbb{R}$$

$$\text{Dominio}(f \circ g) = \mathbb{R}$$

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- $g(3a)$: $g(3a) = 6 - 3(a) = 6 - 3a$
- $g(\frac{3}{a})$: $g(\frac{3}{a}) = 6 - 3(\frac{3}{a}) = 6 - \frac{9}{a}$
- $g(\frac{1}{2})$: $g(\frac{1}{2}) = 6 - \frac{3}{2} = \frac{9}{2}$
- $g(\frac{1}{2}x)$: $g(\frac{1}{2}x) = 6 - 3(\frac{1}{2}x) = 6 - \frac{3}{2}x$
- $g(-2x)$: $g(-2x) = 6 - 3(-2x) = 6 + 6x$
- $g(3 - 2x)$: $g(3 - 2x) = 6 - 3(3 - 2x) = 6 - 9 + 6x = -3 + 6x$
- $g(f(x))$: $g(f(x)) = g(3 - 2x) = 6 - 3(3 - 2x) = 6 - 9 + 6x = -3 + 6x$
- $g(-3x) = 6 - 3(-3x) = 6 + 9x$
- $g(6 - 3x) = 6 - 3(6 - 3x) = 6 - 18 + 9x = -12 + 9x = g(g(x))$

$$\text{Dominio}(g) = \mathbb{R}$$

$$\text{Dominio}(g \circ f) = \mathbb{R}$$

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Para $f(x) = 3 - 2x$; $g(x) = 6 - 3x$, calcular

$$\text{Dominio}(f) = \mathbb{R}$$

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Defina las siguientes funciones y determine el dominio de las funciones resultantes: (a) $f(x^2)$; (b) $[f(x)]^2$; (c) $(f \circ f)(x)$; (d) $(f \circ f)(-x)$, para $f(x) = \frac{1}{x-1}$

$$\bullet f(x^2) = \frac{1}{x^2-1} = \frac{1}{(x-1)(x+1)}$$

$$\bullet [f(x)]^2 = \left[\frac{1}{x-1}\right]^2$$

$$\bullet f(f(x)) = f\left(\frac{1}{x-1}\right) = \frac{1}{\frac{1}{x-1}-1} = \frac{1}{\frac{1}{x-1}-\frac{1}{1}} = \frac{1}{\frac{1-(x-1)}{x-1}} = \frac{1}{\frac{1-x+1}{x-1}} = \frac{1}{\frac{2-x}{x-1}} = \frac{\frac{1}{1}}{\frac{2-x}{x-1}} = \frac{x-1}{2-x}$$

$$\text{Dom}(f) = \mathbb{R} - \{1\}$$

$$\text{Dom}(f) = \mathbb{R} - \{-1, 1\}$$

$$\text{Dom}(f \circ f) = \mathbb{R} - \{2\}$$